TUTORIAL 14

Orthogonality, Best Approximation, and Gram-Schmidt

MOTIVATION AND GUIDING IDEA

An inner product adds geometric measurement to a vector space: we can ask whether two directions are perpendicular, measure size, and decompose a vector relative to a subspace. A different inner product changes the geometry even when the vectors and subspace remain fixed. Begin every computation by identifying the given inner product, and only then apply the formulas.

NOTATION AND CONVENTIONS

- $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is the scalar field; $\overline{z}$ is complex conjugation, with $\overline{z} = z$ over the reals.
- A^T is the transpose and $A^* = \overline{A}^T$ is the conjugate transpose. For column vectors, the standard inner product is $\langle x, y \rangle = x^T \overline{y} = y^* x$.
- $\langle v, w \rangle$ denotes the given inner product, and $\|v\| = \sqrt{\langle v, v \rangle}$ is its induced norm. For a weighted inner product, we sometimes write $\langle v, w \rangle_G$ and $\|v\|_G$.
- $v \perp w$ means $\langle v, w \rangle = 0$; $S^\perp$ is the set of all vectors orthogonal to every element of S.
- δ_{ij} is the Kronecker delta: it is 1 when $i = j$ and 0 otherwise. Thus $\langle e_i, e_j \rangle = \delta_{ij}$ expresses orthonormality.
- P_W is the orthogonal projection map onto W; $P_W(v)$ is the projection of v onto W, and $v - P_W(v)$ is the orthogonal residual.
- $V = W \oplus W^\perp$ means that every $v \in V$ has a unique decomposition $v = w + z$ with $w \in W$ and $z \in W^\perp$.
- $[v]_B$ is the coordinate vector of v relative to the **ordered** basis B; the basis order determines the coordinate order.
- $G \succ 0$ and $G \succeq 0$ denote a positive definite Hermitian matrix and a positive semidefinite Hermitian matrix, respectively.

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DEFINITION 14.1

Let V be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. We use linearity in the first argument and conjugate linearity in the second. Thus, for $u \neq 0$, the projection coefficient onto $\operatorname{span}\{u\}$ is

$$\frac{\langle v, u\rangle}{\langle u, u\rangle},$$

with this order of the arguments in the numerator. The denominator is strictly positive. If V is finite dimensional and $W \leq V$, then $P_W(v)$ is uniquely characterized by the two conditions together:

$$P_W(v) \in W, \qquad v - P_W(v) \in W^{\perp}.$$

If $(e_1, \ldots, e_k)$ is an orthonormal basis of W, then $P_W(v) = \sum_{j=1}^{k} \langle v, e_j\rangle e_j$. For an orthogonal basis that is not normalized, divide each coefficient by $\|e_j\|^2$. Membership alone is insufficient, and orthogonality alone is also insufficient.

MOTIVATION AND GUIDING IDEA

We study how an inner product measures lengths and angles, how to choose an orthonormal basis, and how orthogonal projection decomposes a vector into a component in a subspace and a component perpendicular to it. We connect algebraic computation with geometry and compare different inner products.

PROPOSITION 14.1

Angle in a real space. If $\mathbb{F} = \mathbb{R}$ and $u, v \neq 0$, the number

$$\cos\zeta := \frac{\langle u, v\rangle}{\|u\|\|v\|}$$

satisfies $\cos\zeta \in [-1, 1]$. In particular, there is $\zeta \in [0, \pi]$ for which this equality holds.

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14.1 Orthogonality

DEFINITION 14.2

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Vectors $v, w \in V$ are **orthogonal** if $\langle v, w\rangle = 0$; we write $v \perp w$.

PROPOSITION 14.2

If $v \perp b$ for every $b \in B$, then $v \perp \operatorname{span}(B)$. Indeed, for any linear combination $s = \sum_{j=1}^{k} \alpha_j b_j$,

$$\langle v, s\rangle = \sum_{j=1}^{k} \overline{\alpha_j}\langle v, b_j\rangle = 0.$$

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DEFINITION 14.3

A set $B = \{v_1, \dots, v_n\} \subseteq V$ is **orthogonal** if

$$\langle v_i, v_j \rangle = 0 \qquad \text{for all } i \neq j.$$

If, in addition, $\|v_i\| = 1$ for every i, it is **orthonormal**. In this case,

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If B is also a basis of V, we call it an orthogonal basis or an orthonormal basis, respectively.

RECALL AND TOOLS

Normalization: an orthogonal set can be converted into an orthonormal set: if $\{v_1, \dots, v_n\}$ is orthogonal and contains no zero vector, then $\left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_n}{\|v_n\|} \right\}$ is orthonormal. Indeed, for every $1 \leq i \leq n$, $\left\langle \frac{v_i}{\|v_i\|}, \frac{v_i}{\|v_i\|} \right\rangle = \frac{1}{\|v_i\|^2} \langle v_i, v_i \rangle = 1$.

PROPOSITION 14.3

- An **orthogonal** set containing no zero vector is linearly independent.
- An **orthonormal** set is linearly independent. (Since each of its elements has norm 1, it contains no zero vector.)
- The converse fails: for example, $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ with the standard inner product is linearly independent but is neither orthogonal nor orthonormal.

EXERCISE 14.1

Suppose $u, v \in \mathbb{R}^n$ and $a = 4v - 3u$, $b = u - v$ are orthonormal. Please compute $\|u\|$, $\|v\|$, and $\langle u, v \rangle$, and please prove that u, v are linearly independent.

WORKED SOLUTION

From $u = v + b$ we obtain $a = v - 3b$, hence

$$v = a + 3b, \qquad u = a + 4b.$$

Using $\|a\| = \|b\| = 1$ and $\langle a, b \rangle = 0$ gives

$$\|v\|^2 = 1 + 9 = 10, \qquad \|u\|^2 = 1 + 16 = 17, \qquad \langle u, v \rangle = 1 + 12 = 13.$$

Thus $\|v\| = \sqrt{10}$ and $\|u\| = \sqrt{17}$. If $\alpha u + \beta v = 0$, then

$$(\alpha + \beta)a + (4\alpha + 3\beta)b = 0.$$

Independence of a, b gives $\alpha + \beta = 4\alpha + 3\beta = 0$. Subtracting three times the first equation from the second gives $\alpha = 0$, then $\beta = 0$.

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14.2 Coordinates in an Orthogonal or Orthonormal Basis

DEFINITION 14.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $B = (b_1, \ldots, b_n)$ be an ordered basis of V.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors: $v = \alpha_1 b_1 + \cdots + \alpha_n b_n$.
If B is an **orthogonal** basis, its coefficients are given by $\alpha_i = \frac{\langle v, b_i\rangle}{\|b_i\|^2}$.
If B is an **orthonormal** basis, then $\|b_i\|^2 = 1$, so $\alpha_i = \langle v, b_i\rangle$. Thus the coordinate vector relative to B is

$$[v]_B = \begin{pmatrix} \langle v, b_1\rangle \\ \vdots \\ \langle v, b_n\rangle \end{pmatrix}$$

EXERCISE 14.2

In $\mathbb{C}^3$ with $\langle x, y\rangle = x^T\overline{y}$, let

$$b_1 = \tfrac{1}{13}(5, 12i, 0)^T, \quad b_2 = \tfrac{1}{13}(12, -5i, 0)^T, \quad b_3 = (0, 0, 1)^T.$$

Please show that $B = (b_1, b_2, b_3)$ is an orthonormal basis and please find $[v]_B$ for $v = (-2, 3i, 4)^T$.

WORKED SOLUTION

The first two squared norms are $(25 + 144)/169 = 1$, and the third is 1. Also,

$$\langle b_1, b_2\rangle = \frac{5 \cdot 12 + (12i)(5i)}{169} = 0.$$

The products with b_3 vanish, so these three vectors form an orthonormal basis. With linearity in the first slot, the coordinates are

$$\langle v, b_1\rangle = \frac{-10 + (3i)(-12i)}{13} = 2, \qquad \langle v, b_2\rangle = \frac{-24 + (3i)(5i)}{13} = -3, \qquad \langle v, b_3\rangle = 4.$$

Thus $[v]_B = (2, -3, 4)^T$. Reconstruction gives

$$2b_1 - 3b_2 + 4b_3 = \left(\frac{10 - 36}{13}, \frac{24i + 15i}{13}, 4\right)^T = (-2, 3i, 4)^T.$$

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14.3 Orthogonal Complements and Orthogonal Projections

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DEFINITION 14.5

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. For $S \subseteq V$, define the **orthogonal complement** of S by

$$S^{\perp} := \{\, v \in V : \ \forall s \in S,\ \langle v, s \rangle = 0 \,\}.$$

In particular, if $W \subseteq V$ is a subspace, then $W^{\perp}$ is the **orthogonal complement** of W.

DEFINITION 14.6

Let $U, W \subseteq V$ be subspaces. We say that U and W are **orthogonal** (written $U \perp W$) if

$$\forall u \in U \, \forall w \in W \qquad \langle u, w \rangle = 0.$$

PROPOSITION 14.4

Let $S \subseteq V$, and let $U, W \subseteq V$ be subspaces. The following statements hold:

1. $S^{\perp}$ is a subspace of V.
2. If $S \subseteq T \subseteq V$, then $T^{\perp} \subseteq S^{\perp}$.
3. $(\operatorname{span} S)^{\perp} = S^{\perp}$.
4. $W \cap W^{\perp} = \{0\}$.
5. $U \perp W \iff U \subseteq W^{\perp} \iff W \subseteq U^{\perp}$.
6. $(U + W)^{\perp} = U^{\perp} \cap W^{\perp}$.
7. If V is finite dimensional, then $(U \cap W)^{\perp} = U^{\perp} + W^{\perp}$.
8. If $U \perp W$, then $U + W = U \oplus W$.
9. $W \subseteq (W^{\perp})^{\perp}$. If V is finite dimensional, then $(W^{\perp})^{\perp} = W$.

RECALL AND TOOLS

For all projection results below, assume that V is finite dimensional. In infinite dimensions, closedness of W alone is insufficient in a general inner product space: the projection theorem holds when V is complete in the induced norm, that is, a Hilbert space, and W is closed. Alternatively, it suffices that W be finite dimensional, without assuming completeness of V.

DEFINITION 14.7

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The **orthogonal projection onto** W is the map $P_W : V \to W$ defined by requiring that, for every $v \in V$, $P_W(v) \in W$ be the unique vector satisfying

$$v - P_W(v) \in W^{\perp}.$$

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PROPOSITION 14.5

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The following hold:

1. For every $v \in V$, there are unique $w \in W$ and $z \in W^\perp$ such that
$$v = w + z, \qquad P_W(v) = w.$$
In particular, $V = W \oplus W^\perp$.
2. P_W is linear.
3. $$P_W^2 = P_W, \qquad \mathrm{Im}(P_W) = W, \qquad \ker(P_W) = W^\perp.$$
4. For every $v \in V$,
$$P_W(v) \perp \big(v - P_W(v)\big), \qquad \|v\|^2 = \|P_W(v)\|^2 + \|v - P_W(v)\|^2.$$
5. For all $u, v \in V$,
$$\langle P_W(u), v\rangle = \langle u, P_W(v)\rangle.$$
6. For every $v \in V$, $\|P_W(v)\| \le \|v\|$.

RECALL AND TOOLS

If $\dim V < \infty$ and $\{e_1, \ldots, e_k\}$ is an orthonormal basis of W, then for every $v \in V$,

$$P_W(v) = \sum_{i=1}^{k} \langle v, e_i\rangle\, e_i,$$

and, in particular,

$$W^\perp = \{\, v \in V : \ \langle v, e_i\rangle = 0 \ \forall i = 1, \ldots, k \,\}.$$

PROPOSITION 14.6

If $\dim V < \infty$ and $W \subseteq V$ is a subspace, then

$$\dim W + \dim W^\perp = \dim V.$$

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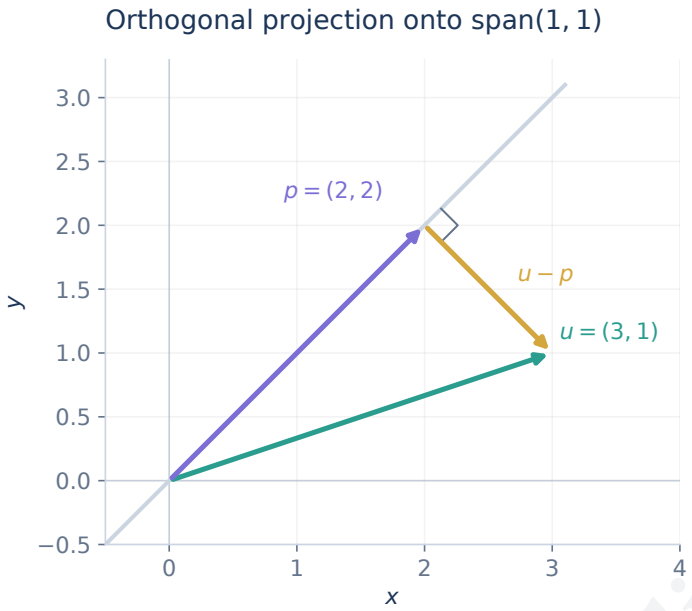


Figure 14.1. Orthogonal projection decomposes a vector into a component in the subspace and a perpendicular residual. The two components satisfy the Pythagorean identity.

EXERCISE 14.3

Let $y = (0, 10)^T$, $u = (3, 1)^T$, and $W = \text{span}\{u\}$. For $\langle x, z\rangle_M = x^T M z$ with $M = \text{diag}(1, 9)$, please compute $P_W^M(y)$ and please check the residual's orthogonality. Please compare with the standard inner product.

WORKED SOLUTION

The matrix M is positive definite because $x^T M x = x_1^2 + 9x_2^2 > 0$ for $x \neq 0$. The projection coefficient is

$$t_M = \frac{y^T M u}{u^T M u} = \frac{0 \cdot 3 + 9 \cdot 10}{3^2 + 9} = 5.$$

Thus $P_W^M(y) = (15, 5)^T$ and $r_M = (-15, 5)^T$. Indeed,

$$r_M^T M u = -15 \cdot 3 + 9 \cdot 5 = 0.$$

For the standard inner product,

$$t_I = \frac{y^T u}{u^T u} = \frac{10}{10} = 1, \qquad P_W^I(y) = (3, 1)^T, \qquad r_I = (-3, 9)^T.$$

Again $r_I^T u = -3 \cdot 3 + 9 = 0$. In each case the projection belongs to W and the residual is orthogonal to W for the chosen inner product.

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Additional practice

EXERCISE 14.4 FOUNDATION

On $\mathbb{R}^2$, let $\langle x, y\rangle_G = 3x_1y_1 + x_2y_2$. Let $u = (1,1)^T$, $v = (2,0)^T$, and $W = \operatorname{span}\{u\}$. Please compute $p = P_W(v)$ and $r = v - p$, and please check orthogonality and the Pythagorean identity.

WORKED SOLUTION

The matrix $G = \operatorname{diag}(3,1)$ is positive definite. We have

$$\langle v, u\rangle_G = 6, \qquad \langle u, u\rangle_G = 4,$$

so

$$p = \tfrac{3}{2}u = (3/2, 3/2)^T, \qquad r = (1/2, -3/2)^T.$$

We have $p \in W$ and $\langle r, u\rangle_G = 3/2 - 3/2 = 0$. Finally,

$$\|v\|_G^2 = 12, \quad \|p\|_G^2 = 3 \cdot \tfrac{9}{4} + \tfrac{9}{4} = 9, \quad \|r\|_G^2 = 3 \cdot \tfrac{1}{4} + \tfrac{9}{4} = 3,$$

and indeed $12 = 9 + 3$.

14.4 Best Approximation

NOTATION AND CONVENTIONS

Let V be an inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $W \leq V$. The distance from $v \in V$ to W is

$$d(v, W) = \inf\{\|v - w\| : w \in W\}.$$

The symbol inf means the greatest lower bound. Under the hypotheses below, an element of W attains this distance, so it is a minimum. Over $\mathbb{C}$, the inner product is linear in the first variable.

PROPOSITION 14.7

Suppose W is finite-dimensional. Every $v \in V$ has a unique closest element $p \in W$. It is the orthogonal projection $p = P_W v$, and for every $w \in W$,

$$\|v - w\|^2 = \|v - p\|^2 + \|p - w\|^2.$$

In particular, $d(v, W) = \|v - P_W v\|$.

Proof. Choose an orthonormal basis $e_1, \ldots, e_d$ of W. Such a basis can be obtained by the Gram-Schmidt process below. Define

$$p = \sum_{j=1}^{d} \langle v, e_j\rangle e_j.$$

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Then $p \in W$, and for every k,

$$\langle v - p, e_k \rangle = \langle v, e_k \rangle - \sum_{j=1}^{d} \langle v, e_j \rangle \langle e_j, e_k \rangle = 0.$$

Thus $v - p \perp W$. If $w \in W$, then $p - w \in W$, so

$$\begin{aligned} \|v - w\|^2 &= \|(v - p) + (p - w)\|^2 \\ &= \|v - p\|^2 + \|p - w\|^2 + 2\,\mathrm{Re}\langle v - p, p - w \rangle \\ &= \|v - p\|^2 + \|p - w\|^2. \end{aligned}$$

The second term is nonnegative and vanishes exactly when $w = p$. Hence p is the unique closest element. For $W = \{0\}$, the same calculation uses an empty sum and $p = 0$.

EXAMPLE 14.1

In $\mathbb{R}^3$ with its standard inner product, let

$$v = (3, 0, 2)^T, \qquad W = \{(a, b, a + b)^T : a, b \in \mathbb{R}\}.$$

The vector $n = (-1, -1, 1)^T$ is orthogonal to every element of W, since $-a - b + (a + b) = 0$. The component of v in the direction of n is

$$r = \frac{\langle v, n \rangle}{\langle n, n \rangle} n = \frac{-3 + 0 + 2}{3} n = \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3} \right)^T.$$

Hence

$$p = v - r = \left(\frac{8}{3}, -\frac{1}{3}, \frac{7}{3} \right)^T.$$

Indeed $8/3 - 1/3 = 7/3$, so $p \in W$. Since $v - p = r \perp W$, this is the projection. We have

$$d(v, W)^2 = \|r\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}.$$

For every $w \in W$, the squared distance is $1/3 + \|p - w\|^2$, whose minimum is $1/3$.

EXERCISE 14.5

Let $v_1, \ldots, v_m \in V$, where $m \geq 1$, and let $W \leq V$ be finite-dimensional. Write $\bar{v} = \frac{1}{m} \sum_{j=1}^{m} v_j$. Please find the unique $w \in W$ minimizing

$$\sum_{j=1}^{m} \|v_j - w\|^2.$$

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EXERCISE 14.5 CONTINUED

WORKED SOLUTION

Write $v_j - w = (v_j - \bar{v}) + (\bar{v} - w)$. Expand each squared norm and sum:

$$\sum_{j=1}^{m} \|v_j - w\|^2 = \sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - w\|^2 + 2\,\mathrm{Re}\left\langle \sum_{j=1}^{m} (v_j - \bar{v}), \bar{v} - w \right\rangle.$$

But $\sum_j (v_j - \bar{v}) = \sum_j v_j - m\bar{v} = 0$. Therefore

$$\sum_{j=1}^{m} \|v_j - w\|^2 = \sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - w\|^2.$$

The first sum does not depend on w. By best approximation, the second term has the unique minimizer

$$w = P_W \bar{v}.$$

The minimum value is

$$\sum_{j=1}^{m} \|v_j - \bar{v}\|^2 + m\|\bar{v} - P_W \bar{v}\|^2.$$

EXERCISE 14.6

Let V be a finite-dimensional inner product space, and let $U, W \leq V$. Please prove that

$$d(x, U)^2 + d(x, W)^2 = \|x\|^2 \quad \text{for every } x \in V$$

if and only if $W = U^{\perp}$.

WORKED SOLUTION

First suppose $W = U^{\perp}$. Decompose $x = u + w$ with $u \in U$ and $w \in W$. The components are orthogonal, so

$$d(x, U)^2 + d(x, W)^2 = \|w\|^2 + \|u\|^2 = \|x\|^2.$$

Conversely, assume the identity for every x. For $u \in U$, we have $d(u, U) = 0$, hence $d(u, W)^2 = \|u\|^2$. Pythagoras for projection onto W gives

$$\|u\|^2 = \|P_W u\|^2 + d(u, W)^2.$$

Thus $\|P_W u\|^2 = 0$, so $P_W u = 0$ and $u \perp W$. We have proved $U \perp W$.
If $z \in (U + W)^{\perp}$, both projections of z vanish. The assumed identity gives

$$\|z\|^2 + \|z\|^2 = \|z\|^2,$$

so $z = 0$. Hence $(U + W)^{\perp} = \{0\}$. Finite dimension gives $U + W = V$. Since $U \perp W$, this is the orthogonal decomposition of V, and therefore $W = U^{\perp}$.

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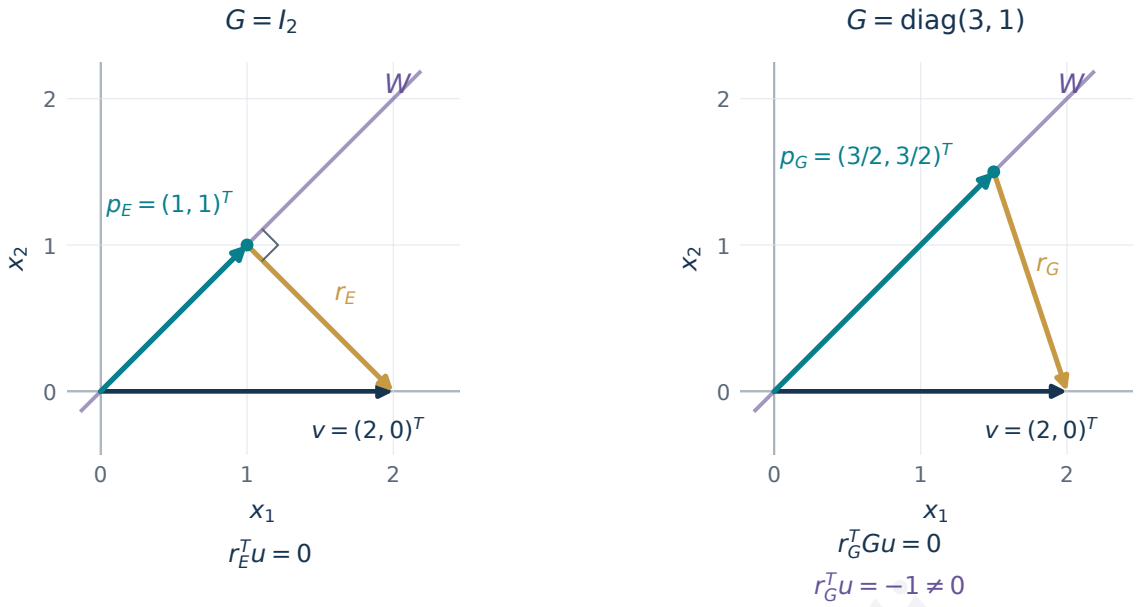


Figure 14.2. Both panels use $v = (2, 0)^T$ and $W = \mathrm{span}\{u\}$, where $u = (1, 1)^T$. On the left, the standard inner product gives $p_E = (1, 1)^T$; on the right, $\langle x, y \rangle_G = x^T G y$ with $G = \mathrm{diag}(3, 1)$ gives $p_G = (3/2, 3/2)^T$. Gold arrows are the residuals. On the right, $r_G^T G u = 0$, whereas $r_G^T u = -1 \neq 0$: orthogonality is with respect to the weighted inner product, so no Euclidean right-angle marker appears there.

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EXERCISE 14.7 CHALLENGE

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $U, W \leq V$. Please write $P = P_U$ and $Q = P_W$, viewed as maps from V to itself.

1. Please prove that PQ is the orthogonal projection onto $U \cap W$ if and only if $PQ = QP$.
2. In $\mathbb{R}^2$ with its standard inner product, take $U = \text{span}\{(1,0)^T\}$ and $W = \text{span}\{(1,2)^T\}$. Please compute PQ and please check whether it is an orthogonal projection.

WORKED SOLUTION

1. Use $\langle Px, y\rangle = \langle x, Py\rangle$, valid for every orthogonal projection, and the corresponding identity for Q. First suppose $PQ = QP$, and set $R = PQ$. For each $x \in V$, the vector Rx belongs to U; since $Rx = QPx$, it also belongs to W. For every $z \in U \cap W$, we have $Pz = Qz = z$, so $Rz = z$. Furthermore,

$$\langle Rx, z\rangle = \langle PQx, z\rangle = \langle Qx, Pz\rangle = \langle x, QPz\rangle = \langle x, z\rangle.$$

Thus $x - Rx \perp U \cap W$. The two defining conditions for projection show that $R = P_{U\cap W}$. Conversely, suppose $R = PQ = P_{U\cap W}$. For all $x, y \in V$,

$$\langle x, QPy\rangle = \langle PQx, y\rangle = \langle Rx, y\rangle = \langle x, Ry\rangle = \langle x, PQy\rangle.$$

Hence $\langle x, (QP - PQ)y\rangle = 0$ for all x, y. For fixed y, choose $x = (QP - PQ)y$; positive definiteness gives $(QP - PQ)y = 0$. Therefore $QP = PQ$.

2. Here

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad PQ = \frac{1}{5}\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}.$$

However,

$$(PQ)^2 = \frac{1}{25}\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \neq PQ.$$

Thus PQ is not an orthogonal projection. Here $U \cap W = \{0\}$, whereas $PQ \neq 0$.

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EXERCISE 14.8 CHALLENGE

Let $a_1, \dots, a_m$ lie in an n-dimensional real inner product space, where $n \geq 1$, and assume $\|a_i - a_j\| = 1$ for all $i \neq j$. Please prove that $m \leq n+1$. Please construct an example attaining the bound in an n-dimensional subspace of $\mathbb{R}^{n+1}$.

WORKED SOLUTION

The case $m = 1$ is immediate. Otherwise put $v_i = a_i - a_m$ for $i < m$. Then $\|v_i\| = 1$, and for $i \neq j$,

$$1 = \|v_i - v_j\|^2 = 2 - 2\langle v_i, v_j \rangle.$$

Thus $\langle v_i, v_j \rangle = 1/2$. For real scalars c_i,

$$\left\| \sum_i c_i v_i \right\|^2 = \frac{1}{2} \sum_i c_i^2 + \frac{1}{2} \left(\sum_i c_i \right)^2 .$$

A zero combination therefore has every $c_i = 0$. The $m-1$ vectors are independent, giving $m - 1 \leq n$. For an example, let $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^{n+1}$ and take

$$H = \{x : x_1 + \cdots + x_{n+1} = 0\}, \qquad a_j = \frac{1}{\sqrt{2}} \left(e_j - \frac{1}{n+1} \mathbf{1} \right).$$

The space H has dimension n, and all $n+1$ points lie in it. For $i \neq j$, $\|a_i - a_j\|^2 = \|e_i - e_j\|^2/2 = 1$, so the bound is attained.

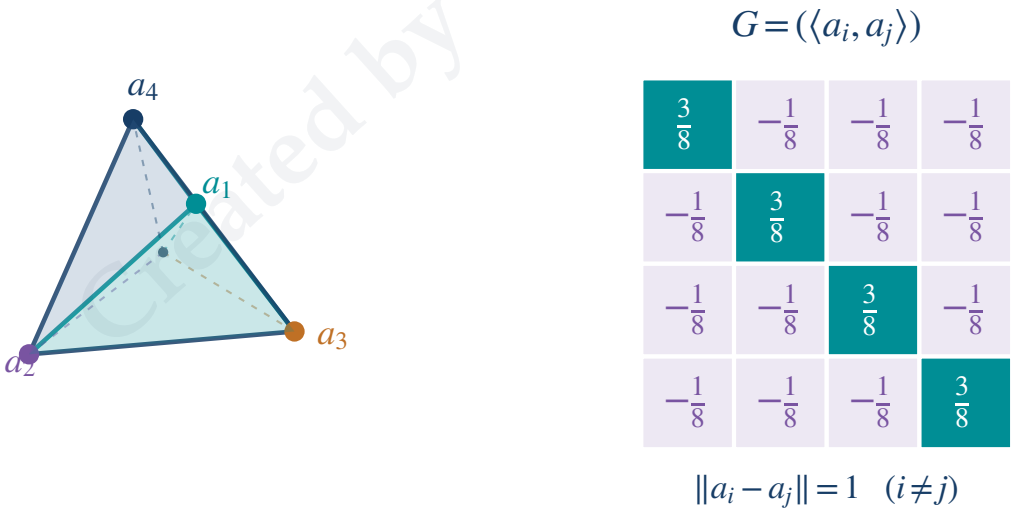


Figure 14.3. In dimension 3, four points can have all pairwise distances equal. The matrix records inner products of the vertices after their centre is placed at zero.

MOTIVATION AND GUIDING IDEA

We construct orthonormal bases by systematically subtracting projections and compute projections in polynomial spaces. We study the conditions for orthogonal diagonalization over the reals and unitary diagonalization over the complex numbers, and use an orthonormal eigenbasis to prove normality.

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14.5 Review and Notation

RECALL AND TOOLS

Throughout this tutorial, the inner product space is over $\mathbb{R}$ or $\mathbb{C}$; in the complex case, our convention is:

1. $\langle x, x\rangle \geq 0$, with $\langle x, x\rangle = 0 \iff x = 0$.
2. $\langle x, y\rangle = \overline{\langle y, x\rangle}$.
3. Linearity in the first argument: $\langle ax + by, z\rangle = a\langle x, z\rangle + b\langle y, z\rangle$.

It follows that the second argument is conjugate linear: $\langle x, ay + bz\rangle = \overline{a}\langle x, y\rangle + \overline{b}\langle x, z\rangle$. This is the same convention as in the preceding tutorial; over $\mathbb{R}$, the two linearity conventions coincide.

DEFINITION **14.8**

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. Vectors $u, v \in V$ are orthogonal if $\langle u, v\rangle = 0$.
2. A list $(e_1, \ldots, e_k)$ is orthonormal if $\langle e_i, e_j\rangle = \delta_{ij}$ for all i, j.
3. For each $x \in V$, define the norm $\|x\| := \sqrt{\langle x, x\rangle}$.

RECALL AND TOOLS

Orthogonal projection onto a line: if $u \neq 0$, the vector

$$\text{proj}_{\text{span}\{u\}}(v) := \frac{\langle v, u\rangle}{\langle u, u\rangle}\, u$$

is the unique vector of the form αu for which $v - \alpha u \perp u$.

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14.6 The Gram-Schmidt Algorithm

We present Gram-Schmidt with immediate normalization at each step, constructing an orthonormal list directly.

Let $v_1, \ldots, v_k \in V$ be linearly independent. Define recursively:

1. Set $w_1 := v_1$, and then $e_1 := \dfrac{w_1}{\|w_1\|}$.
2. For $j = 2, \ldots, k$, first define
$$w_j := v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle\, e_i,$$
and then normalize:
$$e_j := \frac{w_j}{\|w_j\|}.$$

The main conclusions are:

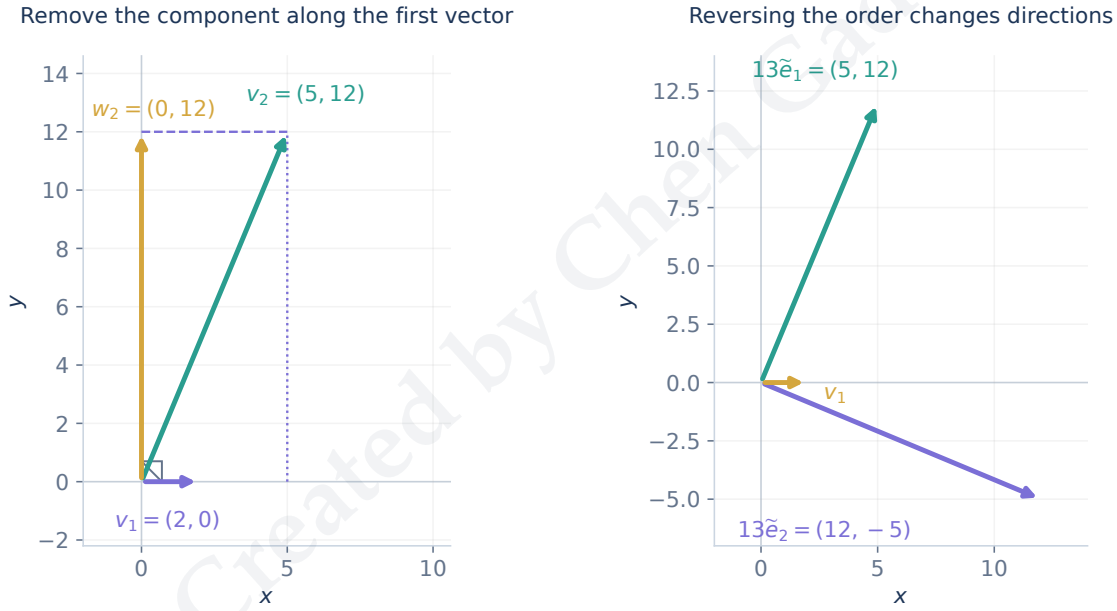
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1. $(e_1, \ldots, e_k)$ is orthonormal.
2. For every m, $\text{span}\{v_1, \ldots, v_m\} = \text{span}\{e_1, \ldots, e_m\}$.

Formal justification: assume inductively that $e_1, \ldots, e_{j-1}$ are orthonormal. For every $r < j$,

$$\begin{aligned}\langle w_j, e_r \rangle &= \left\langle v_j - \sum_{i=1}^{j-1} \langle v_j, e_i \rangle e_i,\ e_r \right\rangle \\ &= \langle v_j, e_r \rangle - \sum_{i=1}^{j-1} \langle v_j, e_i \rangle \langle e_i, e_r \rangle \\ &= \langle v_j, e_r \rangle - \langle v_j, e_r \rangle = 0.\end{aligned}$$

If $w_j = 0$, then v_j belongs to the span of $e_1, \ldots, e_{j-1}$, which by induction equals the span of $v_1, \ldots, v_{j-1}$. This contradicts the linear independence of the input. Thus $w_j \neq 0$, normalization is permitted, and $\|e_j\| = 1$. Also, $w_j \in \text{span}\{v_1, \ldots, v_j\}$, and the subtraction formula implies $v_j \in \text{span}\{e_1, \ldots, e_{j-1}, w_j\}$. Hence the corresponding prefix spans agree at every step.



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Figure 14.4. At each step, subtract from the next vector its components in the directions already constructed. The residual is orthogonal to the preceding directions; normalization produces the next unit vector.

EXERCISE 14.9

Please apply Gram-Schmidt with normalization to

$$v_1 = (5, 12, 0)^T, \qquad v_2 = (1, 5, 0)^T, \qquad v_3 = (2, -3, 4)^T.$$

Please check that the output is orthonormal.

WORKED SOLUTION

First $\|v_1\| = 13$, so $e_1 = (5, 12, 0)^T/13$. We have

$$\langle v_2, e_1 \rangle = \frac{5 + 60}{13} = 5, \qquad w_2 = v_2 - 5e_1 = \tfrac{1}{13}(-12, 5, 0)^T.$$

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EXERCISE 14.9 CONTINUED

The norm of w_2 is 1, so $e_2 = w_2$. At the third step,

$$\langle v_3, e_1 \rangle = \frac{10 - 36}{13} = -2, \qquad \langle v_3, e_2 \rangle = \frac{-24 - 15}{13} = -3.$$

Hence

$$w_3 = v_3 + 2e_1 + 3e_2 = (0, 0, 4)^T, \qquad e_3 = (0, 0, 1)^T.$$

The first two squared norms are $169/169 = 1$, and their inner product is $(-60 + 60)/169 = 0$. Both are orthogonal to e_3, whose norm is also 1.

EXERCISE 14.10

In $\mathbb{C}^2$ with $\langle x, y \rangle = x^T \overline{y}$, let $v_1 = (4, 3i)^T$ and $v_2 = (2, i)^T$. Please apply Gram-Schmidt with normalization.

WORKED SOLUTION

First $\|v_1\|^2 = 16 + 9 = 25$, so $e_1 = (4, 3i)^T/5$. With linearity in the first slot,

$$\langle v_2, e_1 \rangle = 2 \cdot \tfrac{4}{5} + i \cdot \tfrac{-3i}{5} = \tfrac{11}{5}.$$

Thus

$$w_2 = v_2 - \tfrac{11}{5} e_1 = \tfrac{1}{25}(6, -8i)^T, \qquad \|w_2\| = \tfrac{2}{5}, \qquad e_2 = \tfrac{1}{5}(3, -4i)^T.$$

Indeed, $\|e_2\|^2 = (9 + 16)/25 = 1$ and $\langle e_1, e_2 \rangle = (12 + (3i)(4i))/25 = 0$.

EXERCISE 14.11

On $\mathbb{R}[t]_{\leq 2}$, let $\langle p, q \rangle = \int_{-1}^{1} p(t) q(t)\, dt$. Please apply Gram-Schmidt to $(1, t, t^2)$, normalizing at every step.

WORKED SOLUTION

First $\|1\|^2 = 2$, so $e_1 = 1/\sqrt{2}$. Odd functions integrate to zero on a symmetric interval, hence

$$\langle t, e_1 \rangle = 0, \qquad \|t\|^2 = \int_{-1}^{1} t^2 dt = \tfrac{2}{3}, \qquad e_2 = \sqrt{\tfrac{3}{2}}\, t.$$

At the third step,

$$\langle t^2, e_1 \rangle = \tfrac{2}{3\sqrt{2}}, \qquad \langle t^2, e_2 \rangle = 0, \qquad w_3 = t^2 - \tfrac{1}{3}.$$

Its squared norm is

$$\|w_3\|^2 = \int_{-1}^{1} \left(t^4 - \tfrac{2}{3} t^2 + \tfrac{1}{9}\right) dt = \tfrac{2}{5} - \tfrac{4}{9} + \tfrac{2}{9} = \tfrac{8}{45}.$$

Therefore

$$e_3 = \sqrt{\tfrac{45}{8}} \left(t^2 - \tfrac{1}{3}\right) = \sqrt{\tfrac{5}{8}} (3t^2 - 1).$$

These are the first three Legendre polynomials, normalized for this inner product.

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EXERCISE 14.12

In $\mathbb{R}^2$, please compare Gram-Schmidt for the orders (v_1, v_2) and (v_2, v_1), where $v_1 = (2,0)^T$ and $v_2 = (5,12)^T$.

WORKED SOLUTION

The first order gives $e_1 = (1,0)^T$, followed by residual $w_2 = (0,12)^T$ and $e_2 = (0,1)^T$. For the reverse order,

$$\widetilde{e}_1 = \tfrac{1}{13}(5,12)^T, \qquad \langle v_1, \widetilde{e}_1 \rangle = \tfrac{10}{13},$$

so

$$\widetilde{w}_2 = (2,0)^T - \tfrac{10}{169}(5,12)^T = \tfrac{1}{169}(288,-120)^T.$$

Its norm is $24/13$, hence $\widetilde{e}_2 = (12,-5)^T/13$. Both outputs are orthonormal and span the same plane, but their directions differ.

EXERCISE 14.13

Let $V = \mathbb{C}[x]_{\leq 3}$ and define

$$\langle f, g \rangle = \sum_{t \in \{-3,-1,1,3\}} f(t)\overline{g(t)}.$$

Let $W = \mathbb{C}[x]_{\leq 2}$. Please prove that this is an inner product, please find an orthonormal basis of W, and please find $W^{\perp}$.

WORKED SOLUTION

Linearity in the first slot and conjugate symmetry follow from the sum. Also $\langle f, f \rangle = \sum_t |f(t)|^2 \geq 0$. Equality gives four distinct roots, so a polynomial of degree at most 3 must vanish identically. Apply Gram-Schmidt to $(1, x, x^2)$. Symmetry gives

$$\sum_t t = \sum_t t^3 = 0, \qquad \sum_t 1 = 4, \qquad \sum_t t^2 = 20, \qquad \sum_t t^4 = 164.$$

Hence

$$e_1 = \tfrac{1}{2}, \qquad e_2 = \tfrac{x}{\sqrt{20}}, \qquad w_3 = x^2 - \frac{20}{4} = x^2 - 5.$$

The values of w_3 are $4, -4, -4, 4$, so $\|w_3\|^2 = 64$ and $e_3 = (x^2 - 5)/8$. Thus (e_1, e_2, e_3) is an orthonormal basis of W. The polynomial

$$u(x) = 5x^3 - 41x$$

is odd, so it is orthogonal to $1, x^2$. Moreover,

$$\langle u, x \rangle = 5\sum_t t^4 - 41\sum_t t^2 = 5 \cdot 164 - 41 \cdot 20 = 0.$$

It is nonzero and orthogonal to W. Since $\dim W^{\perp} = 4 - 3 = 1$,

$$W^{\perp} = \operatorname{span}\{5x^3 - 41x\}.$$

Its values are $-12, 36, -36, 12$, so $\|u\|^2 = 2880$; division by $24\sqrt{5}$ normalizes it.

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EXERCISE 14.14

Continue with V, W and the preceding inner product. Let P_W be orthogonal projection onto W.

1. Please prove that P_W is linear and $P_W^2 = P_W$.
2. Please compute $P_W(x^3)$ using the orthonormal basis you found.
3. Define $(Tq)(x) = q(x+2) - q(x)$. Please find all $q \in W$ with $Tq \in W^\perp$, and please find the unique solution satisfying $\langle q, 1\rangle = 2$. Please determine whether $\|q\| = 1$ guarantees uniqueness.

WORKED SOLUTION

1. For the orthogonal decomposition $f = w + u$ with $w \in W$, $u \in W^\perp$, we have $P_W f = w$. Combining the decompositions of f, g gives $P_W(\alpha f + \beta g) = \alpha P_W f + \beta P_W g$. Since P_W is the identity on W, we have $P_W^2 = P_W$.
2. Symmetry gives $\langle x^3, e_1\rangle = \langle x^3, e_3\rangle = 0$, while
$$\langle x^3, e_2\rangle = \frac{164}{\sqrt{20}}.$$
Thus
$$P_W(x^3) = \frac{164}{\sqrt{20}}\frac{x}{\sqrt{20}} = \frac{41}{5}x.$$
The residual $x^3 - P_W(x^3) = (5x^3 - 41x)/5 \in W^\perp$ confirms the answer.
3. For $q(x) = ax^2 + bx + c$,
$$(Tq)(x) = a(x+2)^2 + b(x+2) + c - (ax^2 + bx + c) = 4ax + 4a + 2b.$$
Thus $Tq \in W$. Requiring $Tq \in W^\perp$ forces $Tq = 0$, hence $a = b = 0$. All solutions are constants $q = c$. The condition $\langle q, 1\rangle = 4c = 2$ gives $q = 1/2$. In contrast, $\|q\|^2 = 4|c|^2 = 1$ gives $|c| = 1/2$: there are two real solutions and infinitely many complex ones.

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Additional practice

EXERCISE 14.15 FOUNDATION

In $\mathbb{R}^3$ with the standard inner product, consider the list
$$v_1 = (1,1,0)^T, \qquad v_2 = (3,3,0)^T, \qquad v_3 = (1,0,1)^T.$$
Please apply Gram-Schmidt, explicitly handling any zero residual. Please find and please check an orthonormal basis of the input span.

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EXERCISE 14.15 FOUNDATION CONTINUED

WORKED SOLUTION

First, $\|v_1\| = \sqrt{2}$, so

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T.$$

For the second input,

$$\langle v_2, e_1 \rangle = 3\sqrt{2}, \qquad w_2 = v_2 - 3\sqrt{2}\, e_1 = 0.$$

We cannot normalize w_2; we skip it because $v_2 = 3v_1$. For the third input,

$$\langle v_3, e_1 \rangle = \frac{1}{\sqrt{2}}, \qquad w_3 = v_3 - \frac{1}{\sqrt{2}} e_1 = (1/2, -1/2, 1)^T.$$

We have $\|w_3\|^2 = 1/4 + 1/4 + 1 = 3/2$, so the second output vector is

$$e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

A direct check gives

$$\|e_1\|^2 = \frac{1+1}{2} = 1, \qquad \|e_2\|^2 = \frac{1+1+4}{6} = 1, \qquad \langle e_1, e_2 \rangle = \frac{1-1}{\sqrt{12}} = 0.$$

The vectors v_1, v_3 are linearly independent: in $av_1 + bv_3 = 0$, the third coordinate gives $b = 0$, and then $a = 0$. Thus the input span has dimension 2, and (e_1, e_2) is its orthonormal basis. It cannot contain three orthonormal vectors, since those vectors would be linearly independent.

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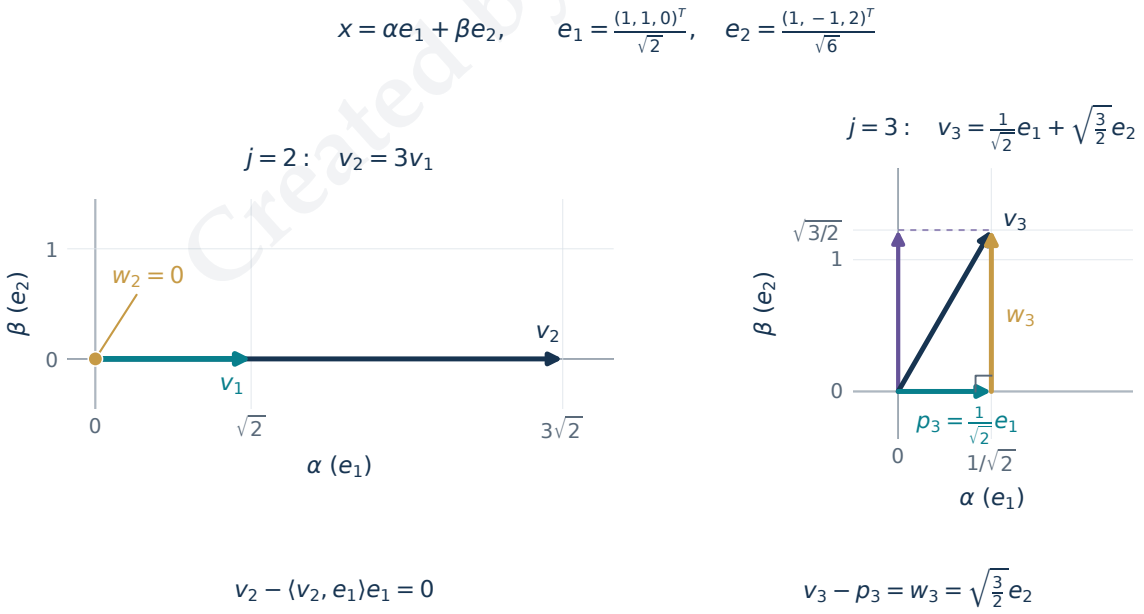


Figure 14.5. The axes α, β are coordinates in the orthonormal basis (e_1, e_2) of the plane $\text{span}\{v_1, v_3\}$: each point represents $x = \alpha e_1 + \beta e_2$. The input coordinates in this chart are $(\sqrt{2}, 0)^T$, $(3\sqrt{2}, 0)^T$, and $(1/\sqrt{2}, \sqrt{3/2})^T$, respectively. On the left, $v_2 = 3v_1$, so the residual $w_2 = 0$ must be skipped. On the right, subtract $p_3 = \langle v_3, e_1 \rangle e_1$ from v_3; the residual $w_3 = (1/2, -1/2, 1)^T = \sqrt{3/2}\, e_2$ is orthogonal to e_1.

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EXERCISE 14.16 EXAM QUESTION

1. Work in $M_{3\times4}(\mathbb{R})$. Define

$$C = \operatorname{diag}(2,0,1), \qquad D = \operatorname{diag}(2,0,0,3),$$

and

$$W = \{X \in M_{3\times4}(\mathbb{R}) : CXD = 0\}.$$

Please prove that W is a subspace of $M_{3\times4}(\mathbb{R})$.

2. On $M_{3\times4}(\mathbb{R})$, define

$$\langle X, Y\rangle = \operatorname{tr}(X^T QY), \qquad Q = \operatorname{diag}(2,3,5).$$

Please prove that this is an inner product. In particular, please verify linearity in the second argument.

3. Now work in $\mathbb{R}^4$ with the standard inner product. Let

$$M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad W = N(M), \qquad v = \begin{pmatrix} 2 \\ -1 \\ 4 \\ 3 \end{pmatrix}.$$

Please compute the orthogonal projection $P_W(v)$.

WORKED SOLUTION

1. Define $L : M_{3\times4}(\mathbb{R}) \longrightarrow M_{3\times4}(\mathbb{R})$ by $L(X) = CXD$. For all $\alpha, \beta \in \mathbb{R}$ and all X_1, X_2,

$$\begin{aligned} L(\alpha X_1 + \beta X_2) &= C(\alpha X_1 + \beta X_2)D \\ &= \alpha CX_1D + \beta CX_2D \\ &= \alpha L(X_1) + \beta L(X_2). \end{aligned}$$

Thus L is linear and $W = \ker L$, so W is a subspace. The condition can also be read directly: multiplication by C zeros the second row, and multiplication by D zeros the second and third columns. Hence

$$CXD = 0 \quad \Longleftrightarrow \quad x_{11} = x_{14} = x_{31} = x_{34} = 0.$$

These are homogeneous linear conditions on the entries of X.

2. Write $Q = \operatorname{diag}(q_1, q_2, q_3)$, where $(q_1, q_2, q_3) = (2, 3, 5)$. If $X = (x_{ij})$ and $Y = (y_{ij})$, computation of the diagonal entries gives

$$(X^T QY)_{jj} = \sum_{i=1}^{3} x_{ij} q_i y_{ij},$$

$$\langle X, Y\rangle = \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} y_{ij}.$$

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EXERCISE 14.16 EXAM QUESTION CONTINUED

For all $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned}\langle X, \alpha Y + \beta Z\rangle &= \sum_{j=1}^{4}\sum_{i=1}^{3} q_i x_{ij}(\alpha y_{ij} + \beta z_{ij}) \\ &= \alpha\langle X, Y\rangle + \beta\langle X, Z\rangle.\end{aligned}$$

This proves linearity in the second argument. Since all entries are real, the same calculation also proves linearity in the first argument. In addition,

$$\langle X, Y\rangle = \sum_{j=1}^{4}\sum_{i=1}^{3} q_i y_{ij} x_{ij} = \langle Y, X\rangle,$$

so symmetry holds. Finally,

$$\langle X, X\rangle = \sum_{j=1}^{4}(2x_{1j}^2 + 3x_{2j}^2 + 5x_{3j}^2) \geq 0.$$

Equality holds if and only if all entries x_{ij} vanish, or equivalently $X = 0$. We have proved linearity, symmetry, and positive definiteness; thus this is a real inner product.

3. Solve $Mx = 0$:

$$x_1 + x_3 = 0, \qquad x_2 + x_4 = 0.$$

Thus

$$W = \operatorname{span}\{w_1, w_2\}, \qquad w_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

The two vectors are linearly independent and orthogonal, and

$$\begin{aligned}\langle w_1, w_2\rangle &= 0, & \|w_1\|^2 &= \|w_2\|^2 = 2, \\ \langle v, w_1\rangle &= 2 - 4 = -2, & \langle v, w_2\rangle &= -1 - 3 = -4.\end{aligned}$$

The projection formula for an orthogonal basis therefore gives

$$\begin{aligned}P_W(v) &= \frac{\langle v, w_1\rangle}{\|w_1\|^2} w_1 + \frac{\langle v, w_2\rangle}{\|w_2\|^2} w_2 \\ &= -w_1 - 2w_2 = \begin{pmatrix} -1 \\ -2 \\ 1 \\ 2 \end{pmatrix}.\end{aligned}$$

As a check, $MP_W(v) = 0$, whereas

$$v - P_W(v) = \begin{pmatrix} 3 \\ 1 \\ 3 \\ 1 \end{pmatrix}$$

is orthogonal to both w_1 and w_2, since $3 - 3 = 0$ and $1 - 1 = 0$. The projection indeed belongs to W, and the residual is orthogonal to all of W.

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EXERCISE 14.17 EXAM QUESTION

Let $V = \mathbb{R}_4[x]$, the space of real polynomials of degree at most 4, including the zero polynomial. Define $L : V \to \mathbb{R}^3$ by

$$L(p) = (p(0), p(1), p'(1)).$$

Equip V with the inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)\,dt + p(0)q(0) + p(1)q(1).$$

Please write $W = \ker L$ and $s(x) = 1 + x + x^2 + x^3 + 2x^4$.

1. Please prove that L is linear.
2. Please compute $\ker L$ and $\operatorname{Im} L$, and please find a basis of each.
3. Please compute the orthogonal projection $P_W(s)$ for the given inner product.

WORKED SOLUTION

1. Let $p, q \in V$ and $a, b \in \mathbb{R}$. By the linearity of evaluation and differentiation,

$$\begin{aligned} L(ap + bq) &= ((ap + bq)(0), (ap + bq)(1), (ap + bq)'(1)) \\ &= (ap(0) + bq(0), ap(1) + bq(1), ap'(1) + bq'(1)) \\ &= aL(p) + bL(q). \end{aligned}$$

   so L is linear.

2. The condition $p \in \ker L$ is equivalent to the three conditions

$$p(0) = 0, \qquad p(1) = 0, \qquad p'(1) = 0.$$

   The first says that x divides p. The last two say that 1 is a root of multiplicity at least 2: if $p(x) = (x-1)q(x)$, then $p'(x) = q(x) + (x-1)q'(x)$, so $p'(1) = q(1)$. Thus $(x-1)^2$ divides p. Since x and $(x-1)^2$ have no common root, and p has degree at most 4,

$$W = \{x(x-1)^2(a + bx) : a, b \in \mathbb{R}\}.$$

   Define

$$w_1 = x(x-1)^2 = x^3 - 2x^2 + x, \qquad w_2 = x^2(x-1)^2 = x^4 - 2x^3 + x^2.$$

   The polynomials w_1, w_2 span W and are linearly independent, for example because they have different degrees. Hence (w_1, w_2) is a basis of W, and $\dim W = 2$.

   To determine the image, exhibit polynomials mapped to the three standard basis vectors:

$$h_1 = (x-1)^2, \qquad h_2 = 2x - x^2, \qquad h_3 = x^2 - x.$$

   A direct calculation gives

$$L(h_1) = (1, 0, 0), \qquad L(h_2) = (0, 1, 0), \qquad L(h_3) = (0, 0, 1).$$

   Thus $\operatorname{Im} L = \mathbb{R}^3$, with the standard basis as a basis of the image. As a dimension check,

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EXERCISE 14.17 EXAM QUESTION CONTINUED

$\dim\ker L + \dim \operatorname{Im} L = 2 + 3 = 5 = \dim V$.

3. Seek $p = P_W(s)$ in the form $p = aw_1 + bw_2$. The definition of orthogonal projection requires $p \in W$ and $s - p \in W^\perp$. Since w_1, w_2 form a basis of W, it suffices to require

$$\langle s - p, w_1\rangle = 0, \qquad \langle s - p, w_2\rangle = 0.$$

Every $w \in W$ satisfies $w(0) = w(1) = 0$, so the endpoint terms in the inner product vanish whenever one factor is w_1 or w_2. First compute the inner products between the basis vectors:

$$\begin{aligned}
\langle w_1, w_1\rangle &= \int_0^1 (t^2 - 4t^3 + 6t^4 - 4t^5 + t^6)\,dt \\
&= \frac{1}{3} - 1 + \frac{6}{5} - \frac{2}{3} + \frac{1}{7} = \frac{1}{105}, \\
\langle w_1, w_2\rangle &= \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7)\,dt \\
&= \frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{1}{280}, \\
\langle w_2, w_2\rangle &= \int_0^1 (t^4 - 4t^5 + 6t^6 - 4t^7 + t^8)\,dt \\
&= \frac{1}{5} - \frac{2}{3} + \frac{6}{7} - \frac{1}{2} + \frac{1}{9} = \frac{1}{630}.
\end{aligned}$$

Multiplying out the polynomials also gives

$$\begin{aligned}
\langle s, w_1\rangle &= \int_0^1 (2t^7 - 3t^6 + t^5 - t^2 + t)\,dt = \frac{2}{8} - \frac{3}{7} + \frac{1}{6} - \frac{1}{3} + \frac{1}{2} = \frac{13}{84}, \\
\langle s, w_2\rangle &= \int_0^1 (2t^8 - 3t^7 + t^6 - t^3 + t^2)\,dt = \frac{2}{9} - \frac{3}{8} + \frac{1}{7} - \frac{1}{4} + \frac{1}{3} = \frac{37}{504}.
\end{aligned}$$

The orthogonality conditions are therefore

$$\begin{pmatrix} 1/105 & 1/280 \\ 1/280 & 1/630 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 13/84 \\ 37/504 \end{pmatrix}.$$

Multiplication by 2520 gives the system

$$24a + 9b = 390, \qquad 9a + 4b = 185.$$

Multiply the first equation by 4 and the second by 9, and subtract:

$$(96a + 36b) - (81a + 36b) = 1560 - 1665 \quad\Longrightarrow\quad 15a = -105.$$

Hence $a = -7$; the second equation gives $-63 + 4b = 185$, so $b = 62$. We conclude that

$$\begin{aligned}
P_W(s) &= -7x(x-1)^2 + 62x^2(x-1)^2 \\
&= 62x^4 - 131x^3 + 76x^2 - 7x.
\end{aligned}$$

The polynomial belongs to W by the factorization in the first line, and the two inner product equations ensure that its difference from s is orthogonal to all of W. These are exactly the two conditions that uniquely characterize the orthogonal projection.

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EXERCISE 14.18 CHALLENGE

Let $v_1, \ldots, v_k$ be independent in a real inner product space, and define

$$u_i = \sum_{j=1}^{i} a_{ji} v_j, \qquad a_{ii} > 0.$$

Please prove that normalized Gram-Schmidt produces the same orthonormal list from $(u_1, \ldots, u_k)$ and $(v_1, \ldots, v_k)$. Please then allow any $a_{ii} \neq 0$ and please describe the relation between the two outputs.

WORKED SOLUTION

For each i, the first i vectors of the two lists span the same subspace. One inclusion follows from the definition; the reverse follows by successively solving for v_i using $a_{ii} \neq 0$. In particular the u_i are independent. Let e_i be the output for the v_i, with residual

$$r_i = v_i - \sum_{j<i} \langle v_i, e_j \rangle e_j, \qquad e_i = r_i / \|r_i\|.$$

Both Gram-Schmidt runs subtract the orthogonal projection onto the same previous span. The terms $a_{ji} v_j$ for $j < i$ lie in that span and disappear from the residual of u_i. Its residual is therefore $a_{ii} r_i$. If $a_{ii} > 0$, normalization gives exactly e_i. For arbitrary nonzero diagonal coefficients, normalization instead gives

$$f_i = \frac{a_{ii} r_i}{|a_{ii}| \|r_i\|} = \frac{a_{ii}}{|a_{ii}|} e_i.$$

Earlier sign changes do not affect orthogonal projections or previous spans, so this relation holds at every step.

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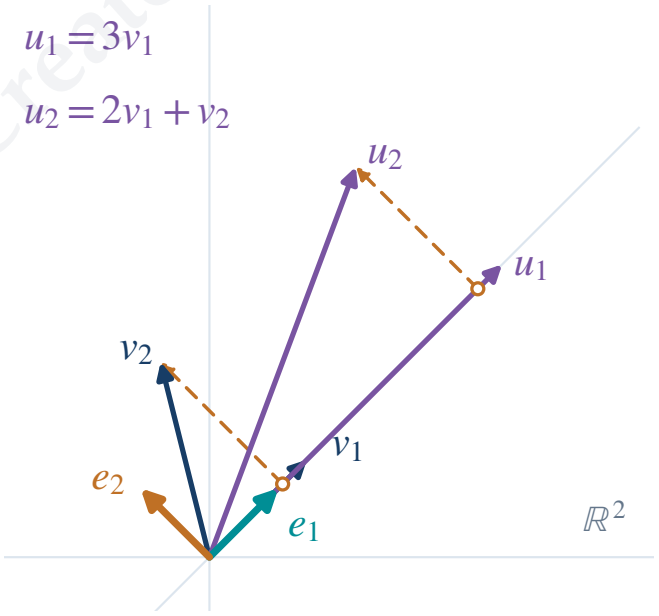


Figure 14.6. Here $u_1 = 3v_1$ and $u_2 = 2v_1 + v_2$. Both lists give the same unit vectors e_1, e_2 under Gram--Schmidt. Dashed segments show the second-step residuals.

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